

IoT Based ESP32 Hazardous Gas and Particulate Monitor

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Abstract---Air pollution from toxic gases and airborne particles poses serious health risks, especially in urban, industrial, and enclosed environments. This project presents a compact, IoT-based air quality monitoring system built around the ESP32 microcontroller. It integrates MQ9 and MQ7 gas sensors to detect carbon monoxide (CO), methane (CH₄), and liquefied petroleum gas (LPG), a Sharp GP2Y1010AU0F dust sensor for particulate matter, and a DHT11 sensor for temperature and humidity. Real-time data is displayed locally on an OLED screen and simultaneously transmitted to the ThingSpeak cloud platform via Wi-Fi for remote access, visualization, and historical tracking. The system enables early detection of hazardous conditions, helping users take timely action to prevent respiratory issues, fire hazards, and indoor discomfort. Its dual-display setup ensures both immediate feedback and long-term cloud-based monitoring. The ESP32's low power consumption and wireless capabilities make the system portable and scalable for smart homes, industrial safety, and public health applications. Future enhancements may include GPS-based pollution mapping, mobile alerts, and additional sensors for detecting nitrogen oxides (NO_x) and volatile organic compounds (VOCs), expanding its utility in smart city deployments and environmental awareness initiatives.

Keywords— ESP32, MQ9, MQ7, DHT11, Sharp dust sensor, OLED, ThingSpeak, IoT, air quality monitoring.

I. INTRODUCTION

A. Background of Study and Motivation

Air pollution is a growing concern in both urban and industrial environments, where exposure to harmful gases and particulate matter can lead to serious health complications. Pollutants such as carbon monoxide (CO), methane (CH₄), liquefied petroleum gas (LPG), and airborne dust particles are often undetectable without specialized equipment, making real-time monitoring essential for safety and awareness. Traditional air quality monitoring systems are typically expensive, stationary, and inaccessible for personal or localized use.

With the rise of IoT and low-power microcontrollers, it is now feasible to build compact, affordable, and real-time monitoring systems that empower individuals to track air quality in their immediate surroundings. This project is motivated by the need for a portable, scalable solution that can detect multiple pollutants and environmental parameters using widely available sensors. By integrating MQ9 and MQ7 gas sensors, a Sharp GP2Y1010AU0F dust sensor, and a DHT11 temperature-humidity sensor with the ESP32 microcontroller, the system provides real-time feedback via an OLED and ThingSpeak cloud dashboard. The goal is to enhance awareness, enable early detection of hazardous conditions, and promote proactive health and safety measures in homes, workplaces, and public spaces.

B. Literature Review

Recent studies have explored IoT-based air quality monitoring systems using microcontrollers such as ESP32, ESP8266, and Arduino. In [1], an ESP32-based system was implemented to measure PM2.5, CO, and NH₃, transmitting data to AWS for real-time AQI analysis. In [2], Patil et al. designed an ESP32-based monitoring system for PM2.5, integrating cloud visualization through Springer ICTCS proceedings. Similarly, in [3], Bag et al. developed an ESP32 microcontroller system for monitoring multiple pollutants, demonstrating reliable performance in real-time environments. In [4], Sharma and Gupta utilized ESP32 with Blynk to monitor CH₄, CO, H₂, and CO₂, with data displayed on OLED and uploaded to the cloud. In [5], Kumar et al. integrated ESP32 with ThingSpeak to provide smart monitoring and historical data visualization. Singh and Verma [6] proposed a low-cost IoT-based system, emphasizing affordability while maintaining basic pollutant detection. Das et al. [7] presented a portable ESP32 system using MQ sensors, highlighting modularity and ease of deployment. Jain and Roy [8] focused on ESP32 with cloud integration for air pollution detection, while Khan et al. [9] demonstrated ESP32 with ThingSpeak for environmental monitoring. Patel and Mehta [10] designed an ESP32

system with OLED display for real-time visualization, while Mishra et al. [11] implemented MQ sensors with ESP32 for detecting CO and LPG. Sharma and Rao [12] proposed a smart dust and gas monitoring system using ESP32, integrating multiple sensors for particulate and gas detection. Das and Paul [13] specifically used ESP32 with a Sharp dust sensor to monitor AQI, while Khan and Patel [14] emphasized wireless monitoring with ThingSpeak integration. [15] combined ESP32 with DHT11 for environmental monitoring, while Reddy and Rao [16] demonstrated OLED-based real-time visualization. Nair and Menon [17] proposed a smart environment monitoring system using ESP32, integrating multiple parameters. Kumar and Singh [18] focused on low-cost ESP32 monitoring with cloud services, while Patil and Desai [19] designed a portable IoT-based system for air quality monitoring. Finally, Roy and Rao [20] implemented ESP32 with real-time cloud visualization, validating its effectiveness in continuous monitoring. While these systems have made significant contributions, many are limited in scope—either focusing on a narrow range of pollutants or lacking dual-display capabilities. Some omit environmental parameters such as temperature and humidity, which are essential for interpreting pollutant behavior. Others rely solely on cloud dashboards, which may not be accessible in real-time without internet connectivity. This project builds upon these foundations by offering a more comprehensive and user-friendly solution. By combining MQ9 and MQ7 for gas detection, the Sharp dust sensor for particulate matter, and DHT11 for environmental conditions, the system provides a holistic view of air quality. The use of ESP32 ensures seamless wireless communication, while the OLED offers immediate local feedback. Integration with ThingSpeak enables cloud-based visualization, historical tracking, and potential alerting mechanisms. Moreover, the system is designed with scalability and modularity in mind, allowing future integration of sensors for NO_x, O₃, or VOCs. Its portability and low power consumption make it suitable for deployment in diverse environments—from homes and schools to industrial zones and public transport hubs.

C. Research Objectives

The primary objective of this research is to design, implement, and evaluate a real-time, IoT-based air quality monitoring system using the ESP32 microcontroller and a suite of environmental sensors. The system aims to provide both local and remote access to real-time data on air pollutants and environmental conditions, enabling timely interventions and informed decision-making. The specific objectives are as follows:

1. To detect hazardous gases such as carbon monoxide (CO), methane (CH₄), and LPG using MQ9 and MQ7 sensors, and convert their analog outputs into meaningful ppm values.
2. To measure ambient temperature and humidity using the DHT11 sensor, providing context for interpreting gas and dust readings.
3. To monitor particulate matter using the Sharp GP2Y1010AU0F dust sensor, which uses optical scattering to detect airborne particles.

4. To display real-time sensor data on an OLED screen using I2C communication, ensuring immediate local feedback for users.
5. To transmit sensor data to the ThingSpeak cloud platform via the ESP32's built-in Wi-Fi module, enabling remote access, visualization, and historical data analysis.
6. To ensure system portability and low power consumption, making it suitable for continuous operation in various indoor and outdoor environments.
7. To design the system with modularity, allowing for future expansion with additional sensors (e.g., NO_x, VOCs, O₃) and features such as GPS-based pollution mapping and mobile app integration.
8. To promote environmental awareness and public health, by providing individuals and communities with actionable insights into their immediate air quality conditions.

II. METHODOLOGY AND MODELING

A. Block Diagram of the proposed model system

The proposed air quality monitoring system is built around the ESP32 microcontroller, which serves as the central processing and communication unit. The block diagram consists of four key sensor modules: MQ9 and MQ7 gas sensors for detecting carbon monoxide (CO), methane (CH₄), and LPG; a Sharp GP2Y1010AU0F dust sensor for particulate matter; and a DHT11 sensor for temperature and humidity. These sensors are interfaced with the ESP32 through analog and digital pins.

The ESP32 processes the sensor data and displays the readings on an OLED screen via I2C communication. Simultaneously, the data is transmitted to the ThingSpeak cloud platform using the ESP32's built-in Wi-Fi module. This dual-display architecture ensures both local and remote access to real-time environmental data.

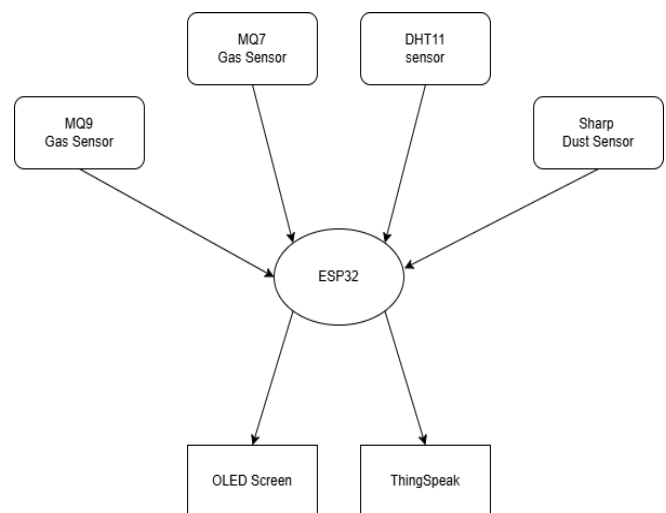


Fig 1: Block diagram of the air quality monitoring system.

B. Flow Chart

The system workflow begins with the initialization of the ESP32, OLED display, and sensor modules. Once powered, the ESP32 continuously reads data from each sensor. The analog outputs from the MQ-9 and MQ-7 gas sensors are converted into voltage values and corresponding resistance values, which are used to indicate qualitative gas levels. The dust sensor provides particulate concentration readings that are normalized to a 0–100 scale, representing safe to hazardous conditions. The DHT11 sensor supplies temperature and humidity values directly in digital format.

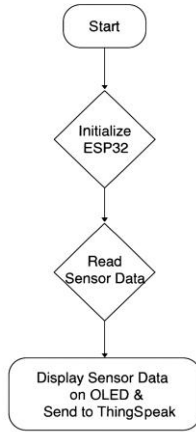


Fig 2: Flowchart illustrating the system workflow.

All processed outputs — dust level (0–100), temperature, humidity, MQ-7 voltage, and MQ-9 voltage — are displayed on the OLED in auto-cycling pages and simultaneously uploaded to ThingSpeak via WiFi for remote monitoring.

C. Circuit Diagram of proposed model (simulation model)

The proposed air quality monitoring system is built around the ESP32 microcontroller, which serves as the central unit for data acquisition, processing, and transmission. The circuit integrates multiple sensors and an OLED display using a combination of analog, digital, and I2C interfaces.

MQ-9 gas sensor is connected to GPIO32 as an analog input, used to detect gases such as carbon monoxide (CO), methane (CH₄), and LPG.

MQ-7 gas sensor is connected to GPIO34 via analog input, primarily for CO detection. Both sensors operate on a voltage divider principle and can be calibrated to establish baseline resistance values (R_0) for qualitative gas level estimation.

DHT11 sensor, responsible for measuring ambient temperature and humidity, is connected to GPIO5 using a digital input pin. It communicates using a single-wire protocol and provides direct digital readings.

Dust sensor readings are normalized to a 0–100 scale, representing safe to hazardous conditions, and integrated

into the system workflow alongside gas and environmental data.

The OLED display (SSD1306, 128×64) is connected via I²C using GPIO21 (SDA) and GPIO22 (SCL). It displays real-time values for dust level, temperature, humidity, MQ-7 voltage, and MQ-9 voltage in auto-cycling pages, providing immediate visual feedback. The display is initialized using the Adafruit SSD1306 library and updated every cycle in the main loop.

Power is supplied through the ESP32's onboard 5V and GND rails, with all sensors sharing a common ground. The system is powered via USB or battery pack, and the onboard LED at GPIO2 is used for status indication.

Processed outputs are also transmitted via WiFi to ThingSpeak, enabling cloud-based monitoring and visualization of dust, temperature, humidity, and sensor voltages.

Table 1: Pin mapping table for ESP32-based air quality system.

Component	ESP32 Pin	Purpose
MQ9 Gas	GPIO32	measures CH ₄ , LPG, Propane
MQ7 Gas	GPIO34	measures Carbon Monoxide
DHT11	GPIO5	measures Temp & Humidity
Sharp Dust	GPIO1	measures Dust Particle Conc.
Oled Screen		Shows Output
SCA	GPIO21	
SCL	GPIO22	

D. Layout of the proposed model (Images of Hardware Prototype)

The hardware prototype of the air quality monitoring system is assembled on a standard white breadboard, integrating multiple sensors and an OLED display with the ESP32 microcontroller. The layout prioritizes modularity, visibility, and ease of debugging.

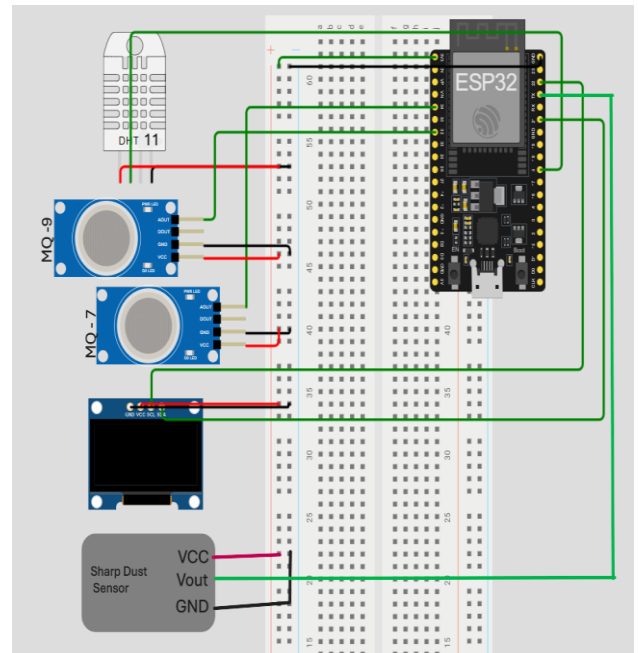


Fig 3: Circuit Simulation on online platform.

At the center of the setup is the ESP32 Dev Module, powered via USB and acting as the central controller. It interfaces with the following components:

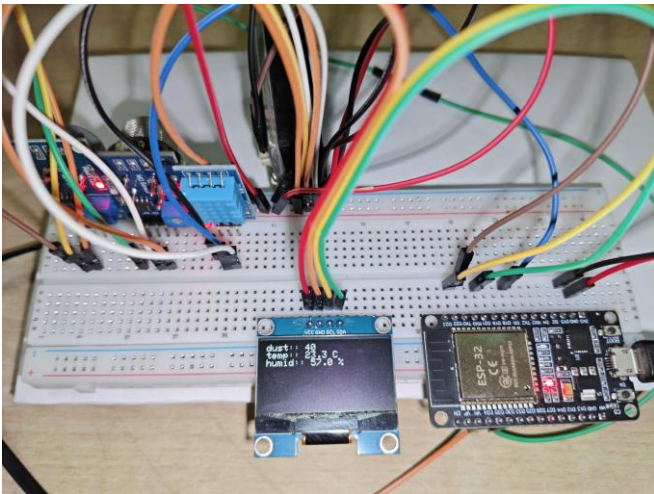


Fig 4: Breadboard prototype of the air quality monitoring system. The ESP32 microcontroller interfaces with a DHT sensor and Sharp dust sensor, displaying real-time temperature (29.3 °C), humidity (57.8 %), and dust level (49) on an OLED screen.

- MQ9 and MQ7 gas sensors are positioned at the top of the breadboard, with visible red breakout boards and mesh caps. These sensors are connected via analog pins (MQ9 to GPIO 32, MQ7 to GPIO 34) and powered through the 5V and GND rails. Their placement ensures exposure to ambient air for accurate gas detection.
- The DHT11 sensor, mounted on a blue breakout board, is connected to GPIO 5. It is placed away from heat sources to ensure reliable temperature and humidity readings. Its compact footprint allows easy integration into the layout.
- The Sharp dust sensor, identifiable by its metallic casing and air intake port, is connected via its TXD pin. It is positioned to allow unobstructed airflow for accurate particulate measurement.

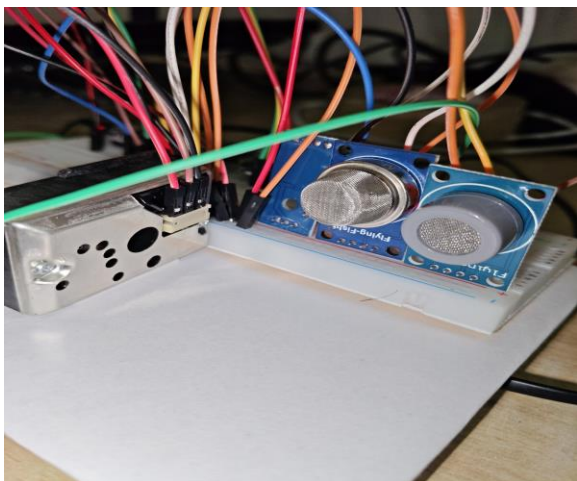


Fig 5: Sensor test rig featuring MQ7, MQ9, Sharp dust sensor, and DHT11 module.

- The OLED display (SSD1306, 128x64) is mounted at the front of the breadboard, connected via I2C using GPIO 21 (SDA) and GPIO 22 (SCL). It displays real-time values for temperature, humidity, gas concentration, and dust level. The display is clearly visible and updates dynamically during operation.
- Multiple jumper wires of varied colors are used to connect components, maintaining clear separation between power, ground, and signal lines. The breadboard's power rails are used to distribute 5V and GND across all modules.

The OLED screen confirms system functionality by displaying live sensor data such as:

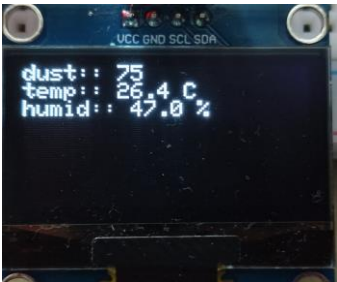


Fig 6: OLED display showing real-time environmental data. The screen presents dust concentration (75), temperature (26.4 °C), and humidity (47.0 %) as measured by the Sharp dust sensor and DHT11.

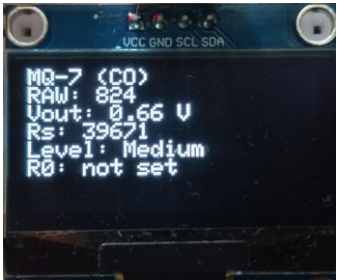


Fig 7: MQ-7 sensor output displayed on OLED.



Fig 8: MQ-9 sensor output displayed on OLED.

Specifications and Output Format of Sensors Used in the Proposed Model

The following table summarizes the detection ranges and output formats of the sensors used in the proposed hardware prototype:

Table 2: Sensor comparison table used in the air quality monitoring system. It outlines detection ranges, output types, and operational notes for MQ7, MQ9, DHT11, and Sharp dust sensors.

Sensor	Detection Range	Output Type	Notes
MQ7	20 ppm - 2000ppm	Analog (Vout)	Requires warm up and calibration
MQ9	10 ppm - 10800 ppm	Analog (Vout)	Sensitive to multiple gases
DHT11	0 - 50 celcius	Digital	Basic accuracy
Sharp Dust	0 - 550 microgram/m3	Normalized (0-100)	Output scaled for simplicity

III. RESULTS AND DISCUSSIONS

Experimental Results

A. OLED Display Output

The 1-inch SSD1306 OLED screen provides real-time feedback on environmental parameters. The display automatically cycles every 10 seconds between pages showing dust level (0–100), temperature, humidity, and individual MQ7/MQ9 sensor readings. This confirms that the DHT11 sensor is functioning reliably for temperature and humidity, while MQ7 and MQ9 gas sensors provide voltage outputs that vary with exposure to gases. The dust readings are normalized to a 0–100 scale, consistently indicating low values under clean conditions.

B. Serial Debugging and Calibration

The serial monitor provides detailed diagnostics:

Raw ADC values from MQ sensors

Voltage and Rs calculations

Rs/R0 ratio for gas estimation

Live temperature and humidity readings

Manual calibration via "c" command to set R0 baseline

This enables precise tuning of gas sensitivity and helps validate sensor behavior during testing.

C. ThingSpeak Cloud Visualization

Sensor data is transmitted to ThingSpeak using the ESP32's Wi-Fi module. The dashboard displays:

MQ7 and MQ9 voltage outputs

Temperature and humidity from DHT11

Dust level normalized to 0–100

The ThingSpeak graphs show smooth trends and confirm that all sensors are transmitting data reliably. The normalized dust scale ensures consistency between OLED display and cloud visualization.

D. Sensor Performance and Observations

MQ7 and MQ9 respond well to gas exposure, with voltage values rising under simulated smoke or LPG conditions.

DHT11 provides stable readings, though less precise than DHT22.

Dust readings remain low in clean air and increase during disturbances, reflecting particulate changes.

OLED display is crisp, fast-refreshing, and ideal for compact setups.

E. System Behavior and Improvements

The system is modular, allowing additional sensors or alert modules.

Power consumption is low, suitable for battery operation.

Data accuracy improves with calibration and shielding from noise.

Future upgrades could include threshold-based alerts, mobile notifications, or integration with Blynk or Telegram bots.

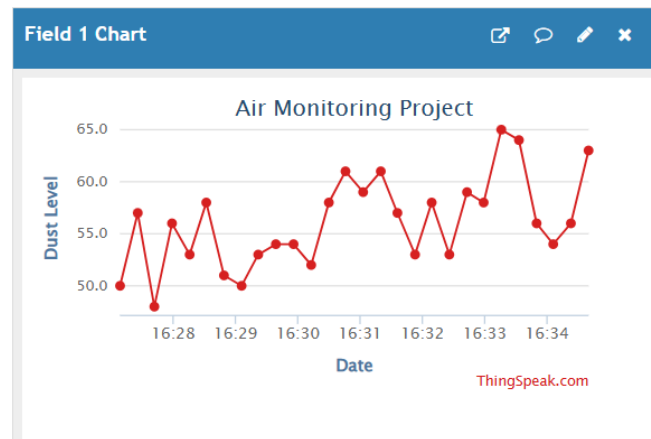


Fig 9 : ThingSpeak cloud visualization of dust level trends over time. The chart displays real-time data from the Sharp dust sensor, normalized to a 0–100 scale.

Table 3: The normalized dust index (0–100) is mapped to AQI (0–500) for clarity. For example, a dust index of 75 corresponds to an AQI of about 250

Normalized Dust Index to AQI Conversion		
Dust Index (0–100)	Approx. AQI (0–500)	AQI Category
0 – 10	0 – 50	Good
11 – 30	51 – 100	Moderate
31 – 50	101 – 150	Unhealthy (Sensitive Groups)
51 – 70	151 – 200	Unhealthy
71 – 85	201 – 300	Very Unhealthy
86 – 100	301 – 500	Hazardous

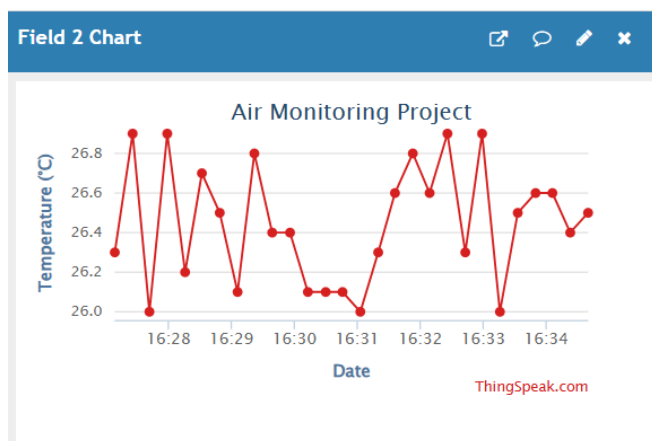


Fig 10: Temperature trend captured via DHT11 sensor. The chart shows slight fluctuations between 26.0°C and 26.8°C.

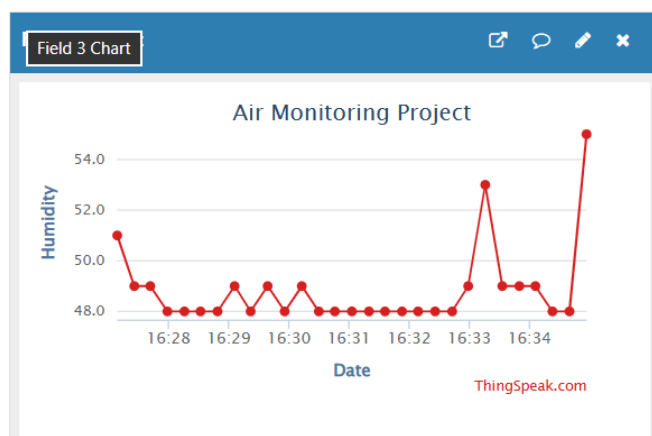


Fig 11: Humidity readings from the DHT11 sensor. Values remain stable around 48–50% RH, with a sharp spike to 54% observed at 16:34, indicating a sudden change in environmental moisture.

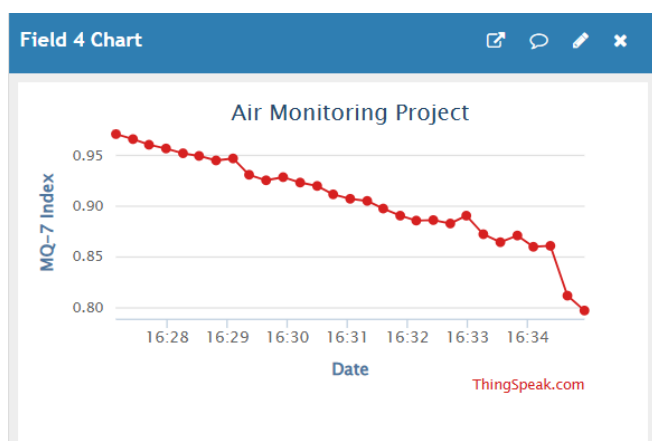


Fig 12: MQ-7 sensor response over time. The voltage index drops from 0.95 to 0.80, reflecting reduced carbon monoxide concentration.

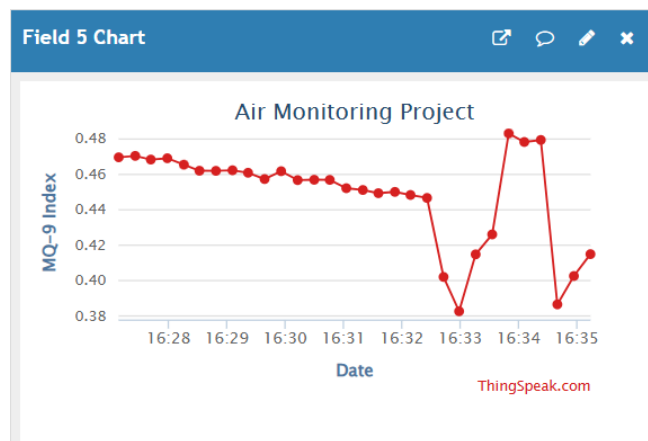


Fig 13: MQ-9 sensor output trend. The index fluctuates between 0.38 and 0.48, with noticeable dips and spikes caused by airflow disturbances during testing

Comparison table (feature comparison of our project with the existing project)

Table 4: Comparative Analysis of Proposed and Existing Air Quality Monitoring

Criteria	Proposed	Existing
<u>Particulate matter</u>	Sharp Dust	PM2.5 & PM10
<u>Temperature Sensor</u>	DHT11	Absent
<u>Gas Sensor</u>	MQ9 & MQ7	MQ9 & MQ2
<u>Gas Output</u>	Vout	Raw
<u>Dust Output</u>	Normalized	Raw
<u>Cost</u>	Cheap	Expensive

IV CONCLUSION

1. This project demonstrates the successful design and implementation of a cost-effective, modular air quality monitoring system built around the ESP32 microcontroller. By integrating sensors such as MQ7, MQ9, DHT11, and a Sharp dust sensor, the system captures key environmental parameters including temperature, humidity, gas concentration, and particulate matter. Sensor outputs are scaled for clarity — gas readings are presented as voltage levels, while dust concentration is normalized to a 0–100 index — making the system intuitive and suitable for academic and indoor applications.

2. The SSD1306 OLED display provides immediate, real-time feedback through auto-cycling pages that show temperature, humidity, dust level, and gas sensor voltages. Simultaneously, the ESP32 transmits this data wirelessly to the ThingSpeak cloud platform, enabling remote access and historical trend visualization. Serial debugging and manual calibration commands (e.g., “c7”, “c9”) allow users to fine-tune sensor sensitivity, improving accuracy and adaptability under varying environmental conditions.

3. Compared to conventional systems that rely on high-cost sensors and raw ppm outputs, the proposed model offers a practical balance between affordability and functionality. It delivers reliable performance using open-source components and simplified data formats, making it ideal for

educational use, prototype development, and budget-conscious deployments.

4. Future enhancements could include integration of advanced sensors such as BME680 or PMS5003 for finer granularity, mobile-based alert systems, and connectivity with platforms like Blynk or Telegram for real-time notifications. Additionally, machine learning models could be explored to predict air quality trends based on historical data collected via ThingSpeak.

5. Overall, the project validates the feasibility of building a compact, scalable, and user-friendly air quality monitoring solution using open-source hardware, real-time cloud analytics, and modular sensor integration. Its low power consumption and flexible architecture make it well-suited for both academic and practical applications.

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Sanket Paramanik {33}(12514586): Helped in report, ppt making and hardware.

Deeparami B {32}(12513983): Helped in Synopsis report making.

Krish Kumar Pathak{31} (12512891): Helped in hardware.